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Efficacy of AI-Driven Evaluation of Exams and Tests

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Abstract — **Across educational systems globally, assessment plays a key role in how student learning is measured and advanced. Every year, assessment volumes grow exponentially, coupled with rising expectations for accuracy and speed. However, grading at such scale brings significant challenges due to evaluator bias, inconsistent application of rubric, logistical inefficiencies, limited feedback quality, and the physical and cognitive toll of working through millions of handwritten answer sheets leaders to evaluator fatigue and errors. Advances in multi-modal AI have opened opportunities to automating much of this work while preserving the standards of trained educators and evaluators. This paper examines how well an AI-driven multimodal system performs when tasked with grading handwritten responses, typed answers, diagrams, mathematical workings and structured reasoning. The system integrates the following key components (1) a vision-language interpretation pipeline for reading a student's work (2) a rubric aligned scoring engine (3) a teacher style visual annotation generator (4) a mastery mapping engine that identifies mastery and trouble spots and (5) an adaptive assessment module that identifies knowledge gaps and recommends personalized follow up questions. The tests measured how closely the system's scores and feedback matched those of experienced human evaluators, and results showed strong agreement between the two. The findings demonstrate that the use of multi modal AI systems in evaluation greatly enhance reliability and effectiveness at scale.**

Keywords—Automated assessment, handwritten evaluation, adaptive testing, mastery mapping, examination economics

I. Introduction

Assessment is an important part of the learning process. It is how a student is provided feedback on their mastery and gaps in learning. At the same time, we cannot ignore the operational challenges involved in the large-scale assessment of exam papers and tests both at school levels and at national or state board levels. Regular and routine processing of substantial numbers of answer scripts in highly populated countries is a challenge in terms of financial and human resource burden. Studies of the National examination system in countries like India have reported evaluation expenditure that are 60-70% of the total expenditure. For example, the CBSE 2022-23 expenditure A budget of ~1500 Crore Rupees comprises a Direct Examination Expense of ~950 Crores [1].

Conventional manual evaluation methods provide numeric scores with minimal qualitative feedback, offering limited actionable guidance for learners. Prevailing reliability concerns due to variability in screening, inconsistent application of rubrics, subjective interpretation of open-ended reasoning, and fatigue during long hours of evaluation sessions compounds these challenges [2].

The economic burden of running large scale examination systems, including security infrastructure, logistics, and evaluator remuneration, increasingly strain educational budgets around the world [3].

The recent developments in multimodal AI, particularly vision-language models (VLMs), deliver a remarkable solution to the challenges that enable machines to interpret and evaluate handwritten answer scripts, diagrams, equations, and problem-solving steps. These advancements present a great opportunity to reform traditional labor-intensive evaluation processes, with a goal to improve consistency, reliability, and feedback quality with a much lower human effort [4].

The application used for this study is an assessment system integrated to AI vision models that accepts images of student handwritten answer sheets, interprets them using AI models, evaluates them against a rubric, generates teacher-style marking and feedback super imposed on the student work. It also generates mastery and trouble spot maps for individualized follow up measures.

This paper presents the details of the AI integrated assessment system, outlines its architecture, the evaluation methodology, and presents illustrative findings demonstrating its potential for deployment in large examination economies.

II. Related Work

A. Automated Assessment

Automated evaluation systems of the past focused on essays, short answers, or multiple-choice questions because they rely on engineered linguistic features [5]. More recent systems and research use transformer-based models for semantic understanding [6] and primarily target typed text, while showing less support for handwritten and diagram-based responses [7]. A comprehensive survey of automated essay scoring [8] reveals the advances and challenges of deploying such systems. Recent systematic literature reviews [8, 9, 10] discuss the evolution of automated essay scoring systems and their efficiencies across diverse educational contexts. Furthermore, recent studies demonstrate continued progress in achieving higher reliability and accuracy [11]. Additional research explores ongoing work on various AI integrated approaches providing much insight into making automated scoring more transparent and interpretable [12,13]. Systematic reviews of LLM-powered automated assessment tools confirm that while high human-AI agreement is achievable, significant challenges remain regarding hallucination, inconsistency, and privacy in large-scale deployments [14].

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B. Vision-Language Models in Education

Vision-Language Models (VLMs) like GPT, Gemini, and similar models have evolved to understand complex inputs, including handwritten math and diagrams [4, 15]. Research indicates that these models achieve near-human consistency in specific applications, such as interpretation, scoring, annotation, and knowledge mapping [13, 15, 16]. Multimodal data fusion in learning analytics [13] demonstrates how combining multiple data modalities can provide richer insights into student learning. The integration of artificial intelligence in multimodal learning analytics [17] represents the frontier of assessment and evaluation technology. Studies examining multimodal learning analytics in relation to student privacy and ethics [18] highlight the importance of responsible implementation. The foundational work on instant feedback and rectification [19] and mastery of foundational concepts before proceeding to advanced topics is key to a successful learning outcome. A recent systematic literature review spanning forty-two peer-reviewed studies further characterizes VLMs as emerging multi-role tools in formal education—functioning as assessors, tutors, and content curators—while highlighting the dominance of the GPT model family across various technology deployment scenarios in education [20].

C. Mastery-Based Feedback and Adaptive Testing

Mastery learning framework prioritizes the identification of conceptual gaps and their remediation [21]. Adaptive testing systems provide follow up assessments by selecting items based on learners' past performance in tests and assessments, by connecting direct evidence extracted from student work to a skill graph. AI-driven assessment systems in higher education [22] represent the convergence of these approaches. Recent research on utilizing artificial intelligence for assessment in higher education [21, 22, 23] demonstrates the practical benefits and implementation challenges. Comprehensive recent reviews [24] outline the opportunities and challenges for broader adoption. Broader systematic reviews of AI-powered adaptive learning confirm that multimodal data integration combining supervised learning, reinforcement learning, and multimodal analytics significantly enhances personalization across diverse learner profiles [25]. Concurrently, scalable assessment techniques driven by LLMs have emerged as a strong alternative to manual processes, enabling personalized, curriculum-aligned feedback at a scale that traditional human-only assessments cannot sustain [26].

D. Economics of Large-Scale Examination Systems

Large-scale examination systems, like those in India, incur substantial direct and indirect costs encompassing logistics, security, evaluator remuneration, and infrastructure [1, 3]. Recent economic audits from 2023 have quantified the fiscal expenditures of national and state level board exams, revealing that evaluator wages, security printing, and surveillance infrastructure consume the majority of operational budgets [1]. Furthermore, recent 2024 studies highlight that the long-run costs of highly competitive exams extend beyond direct fiscal expenditure to include household expenditure on coaching, opportunity costs of test preparation, and mental health impacts [2]. The digital transition from paper-based to on-screen marking represents a strategic cost shift from logistics and transport to IT infrastructure and scanning services, offering immense potential for operational expense reduction in the medium to longer term [4].

III. System Architecture

The application used for this paper consists of five fundamental modules organized in sequential pipelines: (1) an Input Acquisition module where student answers are recorded using photographs, scans, PDFs, or typed submissions (2) a Multimodal Interpretation Engine that reads handwritten text and interprets diagrams. (3) a Rubric Aligned Evaluation Engine that scores responses using both rules based as well as LLM based evaluators, depending on item type. (4) a Teacher Style Annotation Generator that creates visual overlays like highlights, strike throughs, correction arrows, and margin notes. (5) a Mastery Mapping and Adaptive Assessment Module that maps knowledge and skill gaps based on a skill graph and identifies next best questions for personalized learning.

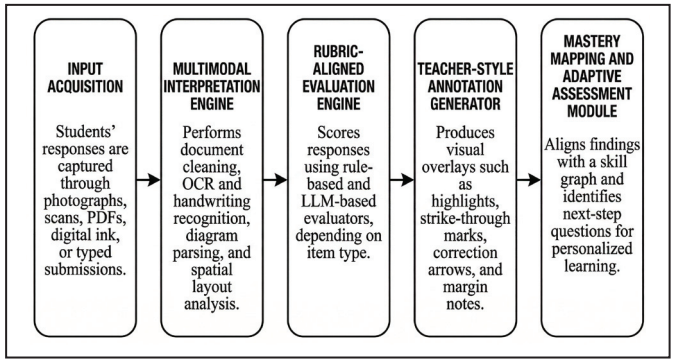


Fig. 1. End to end assessment pipeline from multimodal input acquisition through rubric aligned evaluation, teacher style visual annotation and mastery based adaptive assessment.

A. Input Acquisition and Normalization

When responses are captured under different conditions like varying lighting, camera angles, or uneven scanning, the application implements preprocessing steps like de-skewing, denoising, and segmentation into meaningful regions like answer boxes, diagrams, steps, and graphs. This ensures reliability across different submission environments like PC or tablet camera captures in classrooms and mobile scans at home.

B. Multimodal Interpretation Engine

The interpretation engine utilizes Vision Language Models to derive semantic representations from the responses. This module reads handwritten text, identifies mathematical expressions, understands diagrams and figures, and pieces together the step-by-step logic a student has used to arrive at an answer. What comes out is a structured output — organized like a JSON schema — that captures text segments, mathematical elements, the relationships between ideas, and the physical layout of the response on the page.

C. Evaluation Engine

Depending on the question type, the evaluation engine applies the most appropriate scoring method — exact matching for objective questions, symbolic equivalence checks for mathematical expressions, rubric-driven evaluation for open-ended and descriptive answers, and LLM-assisted scoring where

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rubrics rely on qualitative judgement rather than fixed criteria. For each response evaluated, the engine produces a score based on the provided rubric in numeric or categorical form, a breakdown of the conceptual skills the student has demonstrated, and a record of errors or gaps in their answer.

D. Teacher-Style Visual Annotation Generator

The annotation engine places visual marks directly onto the student's work. Correct steps get highlighted, incorrect ones are underlined or struck through, gaps in reasoning are circled, and arrows link specific steps to explanatory comments in the margin — all tied back to the rubric. The result looks and reads much like feedback a teacher would write by hand, making it straightforward for students to understand exactly where they went right, where they went wrong, and why.

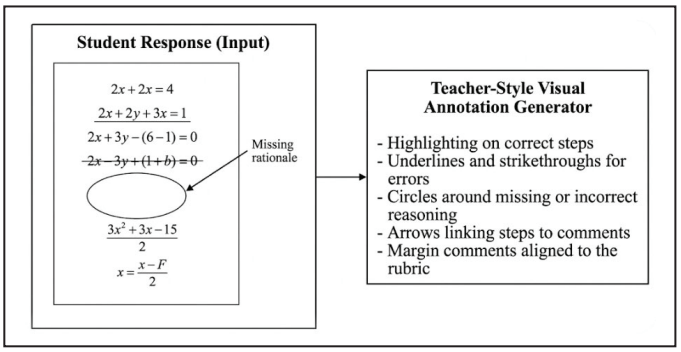


Fig. 2. Teacher style visual annotation generator applying rubric aligned feedback to student responses through highlights, strikethroughs, circles, arrows, and margin comments.

E. Mastery Mapping and Adaptive Assessment

Once the evaluation engine has scored a response, the results are fed into a skill graph, where every node represents a skill, and every connecting edge represents a prerequisite relationship between skills. The mastery engine works through this graph to gauge how well the learner has grasped each concept and to pinpoint where the gaps are. This graph is fed as input to the adaptive assessment module that generates follow-up questions targeting the gaps, adjusting the difficulty of those questions in real time based on how the student is performing.

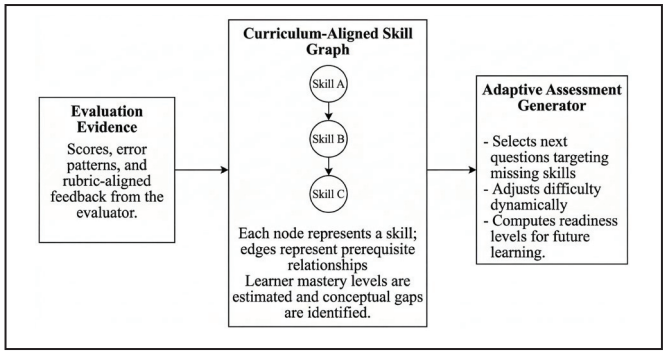


Fig. 3. Mastery mapping and adaptive assessment framework in which evaluation evidence is mapped to a curriculum aligned skill graph to estimate learner mastery, identify gaps, and guide adaptive question selection.

IV. Methodology

A. Dataset

We have utilized a dataset collected from live practice sessions involving Secondary level Grade 10 CBSE students across four core subjects: Mathematics, Science, Social Science, and English. The dataset contains 2091 individual student responses which were complete and suitable for analysis after data validation and cleaning. Each response includes metadata related to subject, difficulty level, question type, maximum marks, student answer, reference answer, AI-generated score, normalized score, and auxiliary properties such as response length.

The dataset comprises 1569 multiple choice questions, 161 assertion and reason questions, 139 very short questions, 106 short questions, 53 long answer questions, and 63 grammar and language questions. The subject distribution consists of 1044 mathematical, 816 social science, 166 science and 65 English questions. Difficulty levels include 1567 beginners, 474 intermediates, and 20 advanced level questions. This distribution ensures robust testing across both objective and subjective formats, reflecting authentic classroom usage.

B. Evaluation Metrics

We have evaluated AI grading performance using: Cohen's Kappa (κ) to measure agreement between AI scores and teacher scores beyond chance, Pearson Correlation (r) to assess linear alignment, Exact Match Accuracy, and Subject-wise Breakdown for reliability analysis stratified by academic subjects.

C. Baselines

In our dataset, we have created a separate column for teachers and student responses that were reviewed and graded by teachers. These scores reflect real classroom evaluations including partial mark allocation, method-based marking, and conceptual coverage assessment. For each response, the dataset contains the "maximum marks" assigned to the question, the "teacher assigned score" (ground truth), and the "AI-generated score" normalized for comparison. By directly comparing AI-generated scores against teacher assigned scores, this study evaluates the true agreement, reliability, and bias characteristics of the automated grading system under realistic educational conditions [1, 3].

V. Results

A. Scoring Reliability

Across all 2091 evaluated responses, the AI grading system demonstrated strong agreement with teacher assigned score grading: Cohen's κ of 0.911, Pearson Correlation (r) of 0.965, and Exact Match Accuracy of 94.98%.

TABLE I. Subject-wise Reliability

Subject	Evaluation Metrics		
	Samples	*Cohen's κ*	*Accuracy*
Mathematics	1044	0.918	95.59%
Social Science	816	0.900	94.36%
Science	166	0.863	92.17%
English	65	1.000	100.00%

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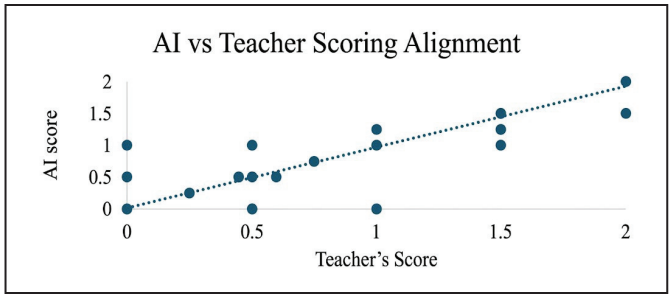


Fig. 4. Alignment between AI generated scores and teacher assigned scores. Each point represents an individual assessment

Mathematics achieved the highest reliability due to deterministic answer structures, while Science showed slightly lower agreement reflecting greater conceptual variability. English showed perfect agreement due to the smaller sample size. These findings align with observed patterns in large scale examination systems where objective questions show higher consistency than subjective evaluations [1, 3].

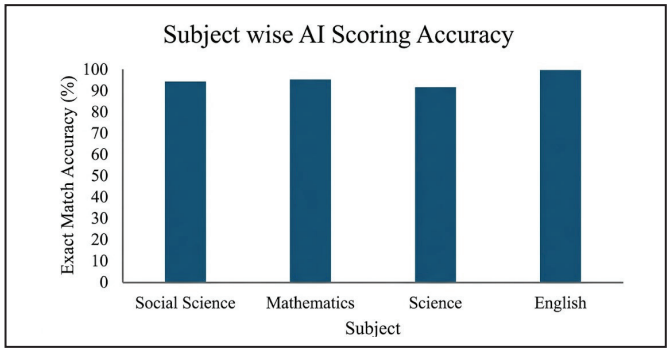


Fig. 5. Subject wise AI scoring accuracy. Exact match accuracy % between AI generated scores and teacher assigned scores across four subjects.

B. *Efficiency*

The efficiency of the grading process was evaluated by comparing human grading time with AI evaluation time. Manual grading durations were approximated based on 30 seconds for multiple choice questions, 60 seconds for very short answers, 120 seconds for short and assertion-reason questions, and 180 seconds for long and language-based responses. Combining these estimates returned a mean human grading time of approximately 51.8 seconds per response. The AI-based evaluation achieved an average processing latency of 2 seconds per response, a twenty-six times reduction in grading processing time. This efficiency gain directly translates to reduced labor costs and faster result processing [1, 3].

C. *Feedback Quality*

To assess the quality of AI generated feedback beyond numerical scoring accuracy, a post evaluation survey was conducted with teachers. Teachers rated AI generated results on a 5-point scale. The aggregated teacher ratings revealed an Annotation clarity of 4.6/5, Correctness of marking rating of 4.7/5 and a Usefulness for student learning rating of 4.8/5. These results suggest that AI assisted grading systems can deliver high quality pedagogical feedback that complement traditional classroom assessment practices [2].

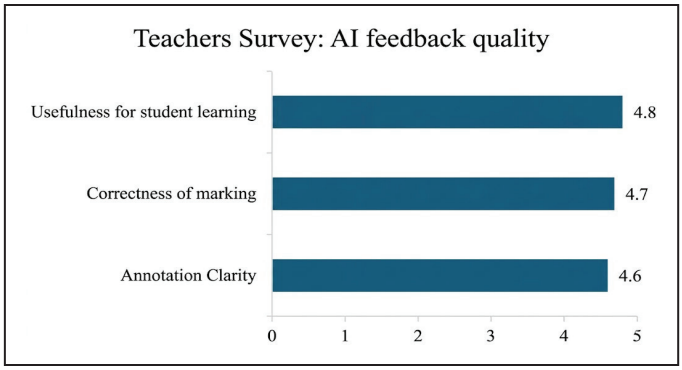


Fig. 6. Teacher perceptions of AI feedback quality. Mean teacher ratings for annotation clarity, correctness of marking, and usefulness for student learning.

D. *Error and Bias Analysis*

We have conducted error and bias analysis by comparing AI generated scores with teacher assigned scores. These results indicate that 94.12% of AI scores were closely aligned with teacher assigned scores, reflecting a high level of agreement. There was a variation from human scoring: 3.11% of responses were marginally overscored, and 2.77% were underscored, suggesting stable evaluative behavior across diverse response types.

A slight positive bias was observed, with the AI tending to award marginally higher scores than human evaluators in some cases. This is worth noting in context — human grading carries its own inconsistencies, shaped by differences between evaluators, whereas the AI applies the same criteria uniformly across every response. Taken together, the error and bias findings suggest that the AI system performs at a level comparable to experienced human evaluators, while delivering greater consistency and reproducibility across the board. That consistency matters most in high-volume settings, where evaluator fatigue and the pressure of large workloads quietly diminish grading reliability over time [3]. Research into how students experience AI grading adds another dimension to this — students tend to see AI evaluators as fairer than teachers, largely because the process feels more transparent to them. Interestingly though, that perception of fairness levels out when students are given clear, detailed explanations of how their work was evaluated [27]. For institutions looking to deploy AI-driven assessment at scale, this points to something practical: how the system communicates its decisions to students matters just as much as how accurately it scores them.

While the AI evaluation system demonstrated high accuracy, it is important to acknowledge its limitations. Across the dataset, 3.11% of responses received marginally higher scores than a human evaluator would have given, while 2.77% were slightly underscored. A closer look at these cases reveals a pattern — most of them involved handwriting so unstructured that the preprocessing steps of de-skewing and denoising could not fully compensate for it, leaving the interpretation engine working from noisier input than it was designed for. Beyond handwriting, the system also met its limits with unconventional diagrams and reasoning that did not follow the patterns on which it was trained. In those instances, the partial marks it assigned occasionally fell short of or exceeded what an expert teacher

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would have awarded, pointing to an area where further refinement is needed.

VI. Conclusion and Future Work

The findings of this study make a clear case for multimodal AI as a practical tool for large-scale assessment. Evaluation cycles that once took trained professionals considerable time and effort can now be completed at a fraction of the cost, with scoring consistency across various subjects and question types. Beyond speed and scale, the system delivers detailed, visual feedback placed directly on the student's work in a form that is immediate and easy to act on.

That said, deploying a system of this kind at scale comes with its own challenges. Educational records are sensitive, and any system handling them at volume must be built around strong data anonymisation and encryption from the ground up — not treated as an afterthought [17]. Rubric alignment needs to be revisited regularly, human oversight must remain part of the workflow, and hallucination in LLM-generated feedback requires active monitoring. Equally important is the question of fairness: scoring patterns need to be watched carefully to ensure that the system does not, over time, disadvantage students from a particular demographic group or those whose handwriting falls outside the patterns the model was trained on.

Looking ahead, connecting this system with learning management platforms already in use by schools and examination boards would be a logical next step — one that could meaningfully accelerate the shift away from labour-intensive evaluation toward a more efficient and scalable model that will deliver a superior and personalized learning experience [4].

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